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HALF YEARLY EXAMINATION - 2025

CLASS : 12

Reg.No

CHEMISTRY

Time : 3.00 Hours

Total Marks : 70

I Choose the correct answer. 15x1=15

1. In the extraction of aluminium from alumina by electrolysis, cryolite is added to
   a) Lower the melting point of alumina
   b) Remove impurities from alumina
   c) Decrease the electrical conductivity
   d) Increase the rate of reduction
2. Oxidadion state of carbon in its hydrides
   a) +4 b) -4 c) +3 d) +2
3. The Oxidation state of halogen in [HXO_2] Halous acid and [HXO_3] Halic acid .....
   a) +3 & +2 b) +2 & +3 c) +3 & +5 d) +5 & +3
4. In acidic medium potassium permaganate oxidizes oxalic acid to .....
   a) Oxalate b) Carbondioxide c) acetate d) acetic acid
5. Which of following complexes having sp^3d^2 hybridisation
   a) $[Ni(CO)_4]$ b) $[FeF_6]^{4-}$ c) $[Co(NH_3)_6]^{2+}$ d) $[Fe(CN)_6]^{2-}$
6. Solid CO_2 is an example of
   a) Covalent solid b) Metallic solid c) Molecular solid d) Ionic solid
7. Assertion : rate of reaction doubles when the concentration of the reactant is doubles if it is a first order reaction.
   Reason : rate constant also doubles.
   a) Both assertion and reason are true and reason is the correct explanation of assertion
   b) Both assertion and reason are true but reason is not the correct explanation of assertion
   c) Assertion is true but reason is false
   d) Both assertion and reason are false
8. Solubility product of $MgSO_4$ is
   a) S^4 b) S^3 c) S^2 d) S^5
9. The number of electrons delivered at the cathode during electrolysis by a current of 1A in 60 seconds is [charge of electron = 1.6×10^{-19}c]
   a) 6.22×10^{23} b) 6.022×10^{20} c) 3.75×10^{20} d) 7.48×10^{23}
10. Which of the following property explain the stability of colloids.
    a) Optical property b) Kinetic property c) Electrical property d) Both (b) & (c)
11. Isopropyl benzene on air oxidation in the presence of dilute acid gives
    a) C_6H_5COOH b) $C_6H_5COCH_3$ c) $C_6H_5COC_6H_5$ d) C_6H_5- OH
12. In the following $HC \equiv CH \xrightarrow[HgSO_4]{H_2SO_4}$ x Product 'X' will not give
    a) Tollens reagent test b) Victor meyer test c) Iodoform test d) Fehling solution test
13. Secondary Nitro alkanes react with nitrous acid to form
    a) Red solution b) Blue solution c) Green solution d) Yellow solution
14. Glucose $\xrightarrow{HCN}$ Product $\xrightarrow{H_2O}$ Product $\xrightarrow[Heat]{HI}$ (A), the compound A is
    a) Heptanoic acid b) 2 - Iodohexane c) Heptane d) Heptanol
15. Which one of the following act as Antioxidants ?
    a) BHA b) PHBY c) HDPE d) PAN

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II Answer any six questions. Question No : 24 is compulsory. 6x2=12

16. Give the limitations of Ellingham diagram.
17. What is Catenation?
18. Write any two biological importance of metal complexes.
19. Classify the following solids.
    (a) Brass (b) NaCl (c) Diamond (d) Solid CO_2
20. Define Solubility product.
21. What is Catalysis? Give example.
22. Write note on Perkin's reaction.
23. Write the note on formation of α - helix.
24. How is aryl halide prepared by using Cu_2Cl_2 / HCl (OR) Cu_2Br_2 / HBr?

III Answer any six questions. Question No : 33 is compulsory. 6x3=18

25. Write a short note on anamolous properties of first element of P - block.
26. Write note on Chromyl chloride test.
27. Define half life of a reaction. Show that for a first order reaction half life is independent of initial concentration.
28. Explain intermediate compound formation theory.
29. State Kohlraush's law.
30. Write short note on transesterification.
31. Write the structure for following compounds.
    (i) D- Glucose (ii) D - Fructose (iii) Zwitter ion structure of alanine.
32. How is terylene prepared?
33. Identify the following in the complex. $[Pt(NO_2)(H_2O)(NH_3)_2]Br$
    (i) Central metal atom / ion (ii) Ligands
    (iii) Oxidation number of the Central metal ion (iv) Co-ordination number
    (v) Types of complex (vi) IUPAC Name

IV. Answer all the questions. 5x5=25

34. a) (i) Explain Magnetic separation. (3)
    (ii) What is Calcination? (2)
    [OR]
    b) (i) What are interhalogen comounds? Give examples. (3)
    (ii) What happens when PCl_5 is heated? (2)
35. a) (i) Write note on Zeolites. (3)
    (ii) Give the uses of Borax. (2)
    [OR]
    b) Discuss briefly assumptions of Valence bond theory.
36. a) Write the difference between Crystalline and Amorphous solids. (5)
    [OR]
    b) (i) Write any two examples for first order reactions. (2)
    (ii) What is petisation? give example. (3).
37. a) Derive Nernst equation. (5)
    [OR]
    b) i) How will you Convert nitrobenzene into (3)
    (1) 1, 3, 5 - trinitro benzene (2) azoxy benzene (3) hydrozobenzene
    ii) What are food preservatives? (2)
38. a) i) Explain Aldol condensation mechanism. (3)
    ii) Write Carbylamine reaction. (2)
    [OR]
    b) Compound (A) is heated with conc. H_2SO_4 at 443 K gives unsaturated hydrocarbon C_2H_4 (B). Compound (B) reacts with Beyer's reagent to give Compound $C_2H_6O_2$ (C), Compound (C) distilled with Conc. H_2SO_4 gives $C_4H_8O_2$ (D). Identity A, B, C & D Explain the reactions. (5)

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